

# Dataset Analysis and Choosing a Dataset

## Part A (Store policy + Customer help desk):

### About the dataset:

A comprehensive policy framework was developed using ChatGPT, drawing references from established policy documents of leading clothing retailers such as H&M, Marks & Spencer (M&S), and Levi's. The formulated document encompasses key operational and customer-centric policies, including exchange, refund, and return procedures, garment recycling initiatives, loyalty and reward programs, as well as gifting policies.

### Scoring Metric (Part A):

#### Retrieval-Augmented Generation Assessment Suite (RAGAS)

RAGAS is a specialized evaluation framework for RAG systems. It goes beyond basic overlap metrics (BLEU, ROUGE) by focusing on how well the generated answer is grounded in retrieved evidence and how relevant / faithful it is.

<Image>

### Scoring Metric (Part B Tentative):

1. In the research papers, models trained would find a compatible top/bottom, but filters were not applied. But in our project filtering on user query is going to be a major part and so we will define a custom scoring metric.
2. Tentative scoring metric (for illustration)

$$\text{score} = 0.2 * \text{color} + 0.2 * \text{material} + 0.2 * \text{gender} + 0.2 * \text{category} + 0.2 * \text{compatibility}$$

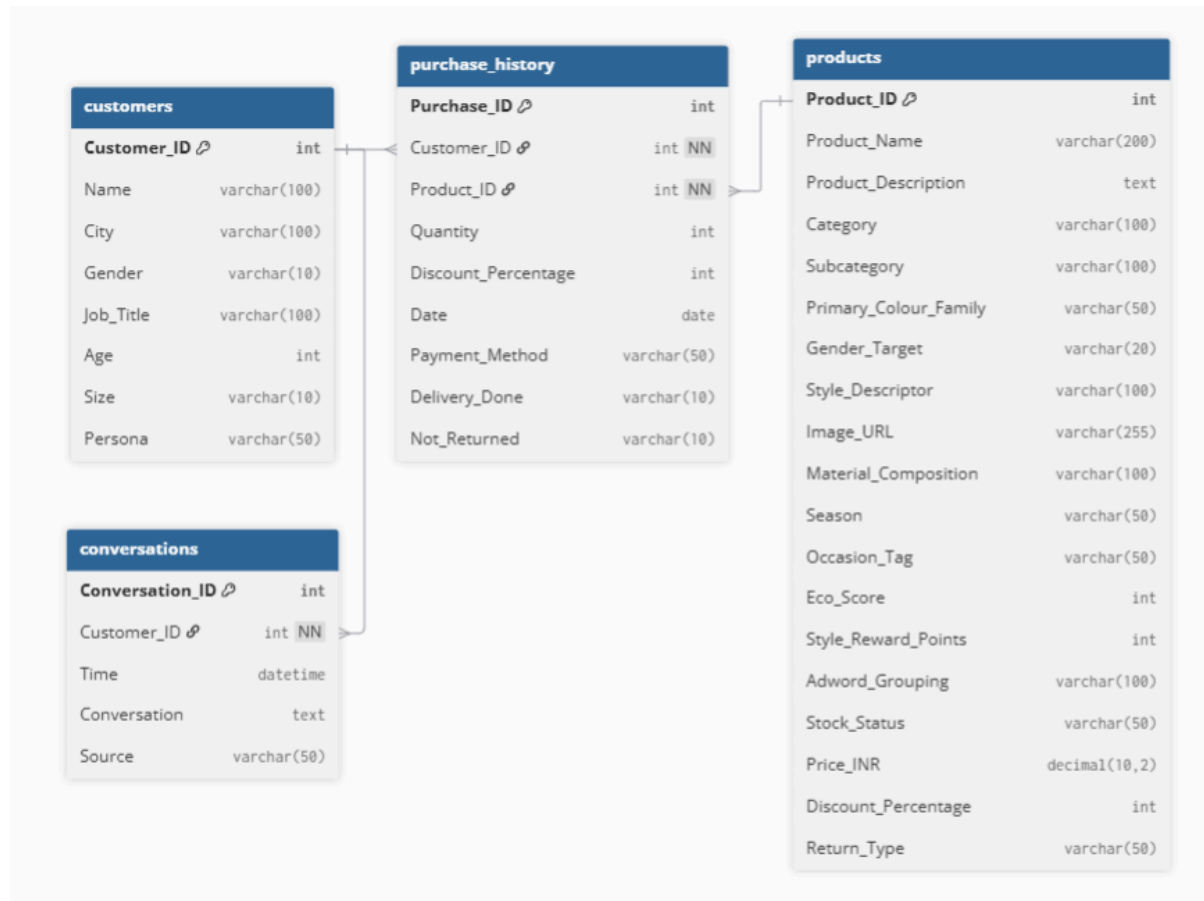
3. Any filter that is not provided in the user query will be ignored during score calculation and we will scale the score based on the filters available.
4. For compatibility we experimented with metrics like color harmony, but if we get access to a trained model from the research sources we have referred to we may use that.

### Model Performance Test (for Baseline and Benchmarks):

1. Initially, we will not give the model much context about the nature of the data and specific instructions which it needs to follow for better search. We will use this as a baseline.

2. Subsequently we will define steps to follow for better search and also will try to increase the llm's understanding of fashion (compatibility text based) and will compare the performance with the baseline.
3. Additionally we will try different embedders and compare the performance.

## Dataset Schema



Product Catalog

| Product Catlog        |                                                       |                                                                                   |
|-----------------------|-------------------------------------------------------|-----------------------------------------------------------------------------------|
| Column Name           | Description                                           | Sample data                                                                       |
| Product id            | Unique product identifier                             | M-DNM-001                                                                         |
| Product_Name          | Short marketing name                                  | Slim Fit Blue Jeans                                                               |
| Product_Description   | Concise but rich product detail (style, fit, comfort, | "Mid-rise slim-fit denim with stretch fabric for all-day comfort."                |
| Category              | Top-level grouping                                    | Men / Women / Teens / Kids / Unisex                                               |
| Subcategory           | Secondary grouping                                    | Denim / Tops / Dresses / Trousers / Jackets / Pyjamas / Swimwear / Glasses / Caps |
| Season                | Target release or style season                        | Summer / Winter / Spring / Autumn / All Season                                    |
| Primary_Colour_Family | Dominant color tone                                   | Blue / Black / White / Beige / Red                                                |
| Gender_Target         | Intended wearer group                                 | Men / Women / Boys / Girls / Unisex                                               |
| Material_Composition  | Top 2 materials                                       | Cotton, Elastane                                                                  |
| Occasion_Tag          | When to wear                                          | Casual / Formal / Party / Beach / Sports / Sleepwear                              |
| Style_Descriptor      | Fashion tone                                          | Streetwear / Classic / Modern / Relaxed / Elegant                                 |
| Eco_Score             | Sustainability rating (A–E)                           | B                                                                                 |
| Style_Reward_Points   | Points earned on purchase                             | 25                                                                                |
| Adword_Grouping       | For search/ad campaigns                               | "Slim Jeans Men"                                                                  |
| Stock_Status          | Inventory state                                       | In Stock / Few Left / Out of Stock                                                |
| Price_INR             | Product price                                         | 2499                                                                              |
| Discount_Percentage   | Discount available                                    | 15                                                                                |
| Return_Type           | Return policy                                         | Returnable / Non-returnable                                                       |
| Image_URL             | Image path or URL                                     | /images/M-DNM-001.jpg                                                             |

| A      | B                                         | C                      | D                  | E                 | F                  | G             | H                | I                | J                   | K      | L            | M         | N                   | O                 | P            | Q         | R                   | S              |
|--------|-------------------------------------------|------------------------|--------------------|-------------------|--------------------|---------------|------------------|------------------|---------------------|--------|--------------|-----------|---------------------|-------------------|--------------|-----------|---------------------|----------------|
| SKU_No | Product_Name                              | Product_Description    | Category           | Subcategory       | Primary_Colour_Fam | Gender_Target | Style_Descriptor | Image_URL        | Material_Compositio | Season | Occasion_Tag | Eco_Score | Style_Reward_Points | Adword_Grouping   | Stock_Status | Price_INR | Discount_Percentage | Return_Type    |
| 1      | 3p fancy tanktop bod                      | Sleeveless bodysuits   | Garment Upper body | Vest top          | White              | Men           | All over pattern | images034034499f | Cotton Blend        | Spring | Sports       | 85        | 40                  | garment_upper_bod | Out of Stock | 4896      | 14                  | 15 days return |
| 2      | Scale tee                                 | Long T-shirt in slab c | Garment Upper body | T-shirt           | Dark Red           | Men           | Solid            | images0705070294 | Cotton Blend        | Spring | Sports       | 91        | 86                  | garment_upper_bod | Low Stock    | 2815      | 44                  | 15 days return |
| 3      | Lina Jumpsuit                             | Jumpsuit in a pattern  | Garment Full body  | Jumpsuit/Playsuit | Black              | Men           | All over pattern | images078078798f | Cotton Blend        | Spring | Formal       | 92        | 43                  | garment_full_body | In Stock     | 3329      | 7                   | 7 days return  |
| 4      | KATE BLOUSE                               | Blouse in a pattern    | Garment Upper body | Blouse            | White              | Men           | All over pattern | images073073280f | Cotton Blend        | Spring | Party        | 99        | 49                  | garment_upper_bod | In Stock     | 1251      | 64                  | 15 days return |
| 5      | Kiev sportswear denim jacket in washed de | Garment Upper body     | Jacket             |                   | Black              | Men           | Denim            | images069069992f | Cotton Blend        | Summer | Party        | 58        | 79                  | garment_upper_bod | Out of Stock | 1577      | 12                  | 7 days return  |
| 6      | LOTTA linen dress                         | Baby Exclusive: One    | Garment Full body  | Dress             | Dark Blue          | Men           | Solid            | images070707039f | Cotton Blend        | Spring | Formal       | 86        | 80                  | garment_full_body | Out of Stock | 2171      | 8                   | No return      |

Customer Schema

| Customer_schema |                                           |                    |
|-----------------|-------------------------------------------|--------------------|
| Column Name     | Description                               | Sample data        |
| Customer ID     | Unique id for each customer               | 2                  |
| Name            | Name of the Customer                      | Joshua Miller      |
| City            | Location of the customer                  | Delhi              |
| Gender          | Gender of the Customer                    | F                  |
| Job Title       | Job title of the customer                 | Records manager    |
| age             | Customer age                              | 25                 |
| size            | Prefferd size of the customer             | XS                 |
| persona         | What type of outfits the customer prefers | Elegant Minimalist |

|   | A           | B                   | C      | D      | E                     | F   | G    | H                    |
|---|-------------|---------------------|--------|--------|-----------------------|-----|------|----------------------|
| 1 | Customer ID | Name                | City   | Gender | Job Title             | age | size | persona              |
| 2 | 1           | Tyler Garcia        | Chenni | F      | Agricultural enginee  | 22  | L    | Trendy Buyer         |
| 3 | 2           | Joshua Miller       | Delhi  | F      | Records manager       | 25  | XS   | Elegant Minimalist   |
| 4 | 3           | Alison Marshall DDS | Chenni | F      |                       | 22  | L    | Comfort Lover        |
| 5 | 4           | Jeffery Acosta      | Delhi  | F      | Proofreader           | 28  | L    | Party Enthusiast     |
| 6 | 5           | Ashley Sanders      | Chenni | M      | Exercise physiologist | 27  | L    | Streetwear Guy       |
| 7 | 6           | Steven Chavez       | Chenni | M      | Technical brewer      | 37  | M    | Minimalist Professor |
| 8 | 7           | Steven Snyder       | Delhi  |        |                       | 24  | L    |                      |

Transactions Schema

| Transactions schema |                                                 |             |  |
|---------------------|-------------------------------------------------|-------------|--|
| Column Name         | Description                                     | Sample data |  |
| customer id         | Unique id for each customer                     | 1           |  |
| product id          | Unique product identifier                       | 111         |  |
| quantity            | Quantity bought by customer                     | 1           |  |
| discount %          | percentage of Discount applied on selling price | 20%         |  |
| date                | Date of purchase                                | 7/9/2024    |  |
| payment method      | Payment method used                             | Cash        |  |
| delivery done       | Delivery status                                 | Delivered   |  |
| returned            | Did customer return the product or not          | No          |  |

| customer id | product id | quantity | discount % | date      | payment meth | delivery done | returned |
|-------------|------------|----------|------------|-----------|--------------|---------------|----------|
| 1           | 17         | 1        | 10%        | 6/3/2024  | Cash         | Delivered     | No       |
| 1           | 47         | 1        | 20%        | 6/15/2024 | Cash         | Delivered     | No       |
| 1           | 78         | 1        | 15%        | 6/21/2024 | Cash         | Delivered     | No       |
| 1           | 73         | 1        | 20%        | 7/1/2024  | Cash         | Delivered     | No       |
| 1           | 111        | 1        | 20%        | 7/9/2024  | Cash         | Delivered     | No       |
| 1           | 21         | 1        | 10%        | 7/12/2024 | Cash         | Delivered     | No       |
| 2           | 6          | 4        | 15%        | 7/20/2024 | Cash         | Delivered     | Yes      |
| 2           | 3          | 1        | 5%         | 7/25/2024 | Credit Card  | Delivered     | No       |
| 2           | 117        | 1        | 25%        | 8/2/2024  | Credit Card  | Delivered     | No       |